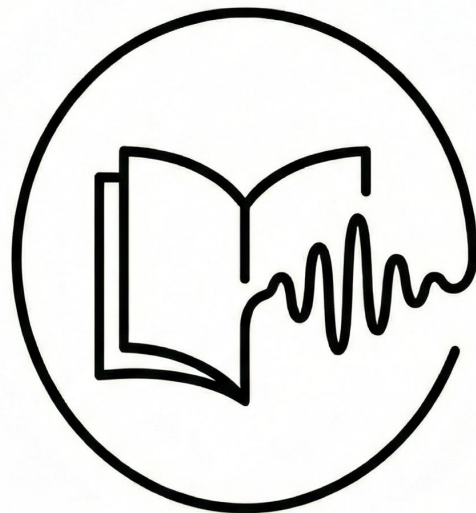




Accessible Academic

Empowering Education Through AI Accessibility

Presented by: Adi Tapiro & Matan Gurfinkel

Mentor: Dror Margalit

Project no.15002930

Entrepreneurship Workshop in Software Development 2025

A Mission Born from Personal Struggle



Adi's Story: Hearing Impairment

"Accessibility shouldn't be a checkbox; it should be a commitment."

Adi Tapiro, faced the daily exhaustion of following muffled lecture recordings without accurate Hebrew captions. This wasn't just a technical flaw; it was a barrier to her basic right to understand.

Accessible Academic was created to ensure no student is left behind in a digital-first world.

Silent Barriers in Modern Education



Accessibility Gap

Students who are hard of hearing struggle with recordings lacking accurate Hebrew captions.



Poor Audio Quality

Large hall acoustics and low-quality mics make lectures muffled even for general students.



Information Overload

Massive amounts of raw video content are difficult to navigate and summarize manually.

Our Solution Bridges the Inclusive Gap



Audio Rescue

Intelligent gain boosting and normalization to transform muffled hall acoustics into clear speech.



Accurate Hebrew STT

Specialized AI pipeline for Hebrew academic terminology, delivering synchronized word-level captions.



Smart Study Suite

Automatic extraction of key topics, summaries, and interactive flashcards to enhance retention.

A Layered AI Pipeline Architecture



Audio Enhancement

FFmpeg (Microosoft)
-driven Gain Normalization



Raw Transcription

High-precision Hebrew
Speech-to-Text



Logical Refinement

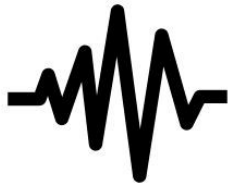
GPT-4o powered Academic
Grade Editing



Knowledge Synthesis

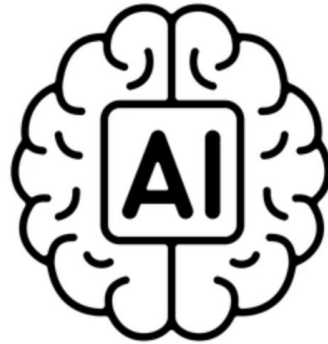
Extraction of Summaries &
Flashcards

Leveraging Best-of-Breed AI Engines



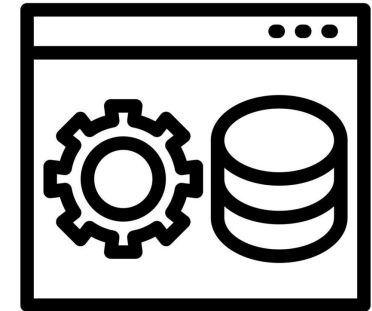
AssemblyAI

Best-in-class Hebrew STT for high accuracy in technical academic contexts.



OpenAI GPT-4o

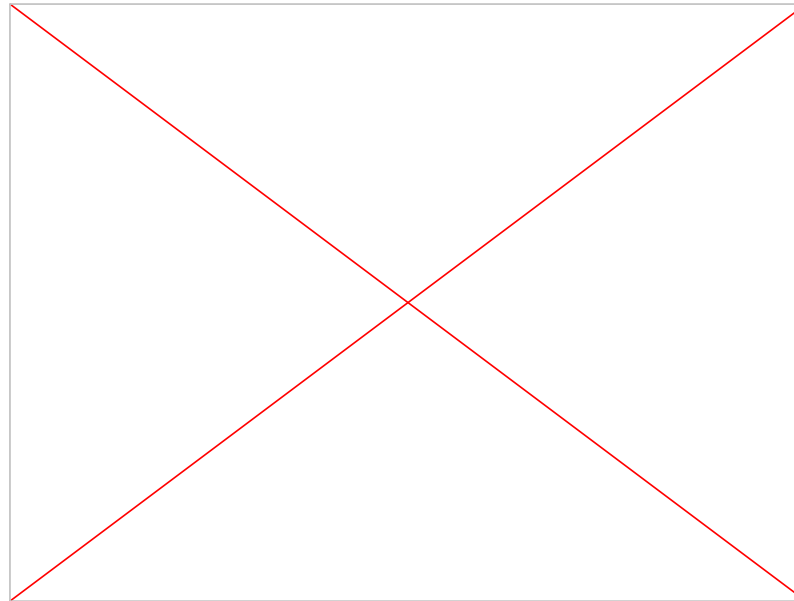
Advanced for understanding transcript, summarization, and interactive tools.



Python / FastAPI

Robust, asynchronous backend for handling high-volume media processing.

Functional MVP Delivering Real Impact



Our Pilot MVP Includes:

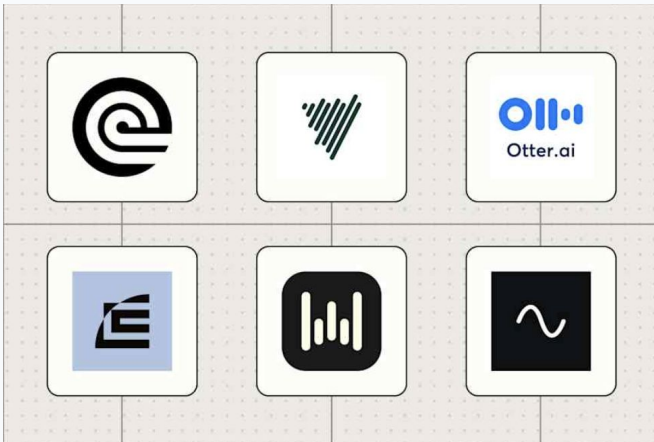
Full Hebrew/English UI Toggle | Show/Hide Synchronized Captions | Deep-Dive Summaries |
Interactive Flashcards



Market Landscape: Why Our Solution Stands Out

Generic STT Tools

- Otter.ai / Zoom Transcripts and more....
- Basic accuracy in Hebrew
- No academic context



Manual Services

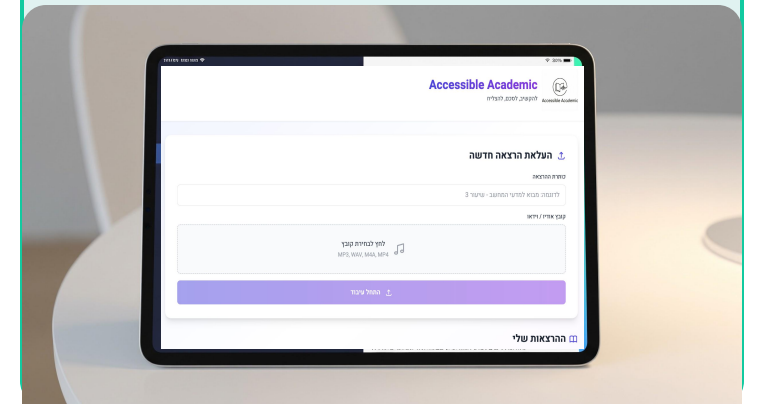
- Human transcribers
- Prohibitively expensive
- Turnaround time: 24-48h



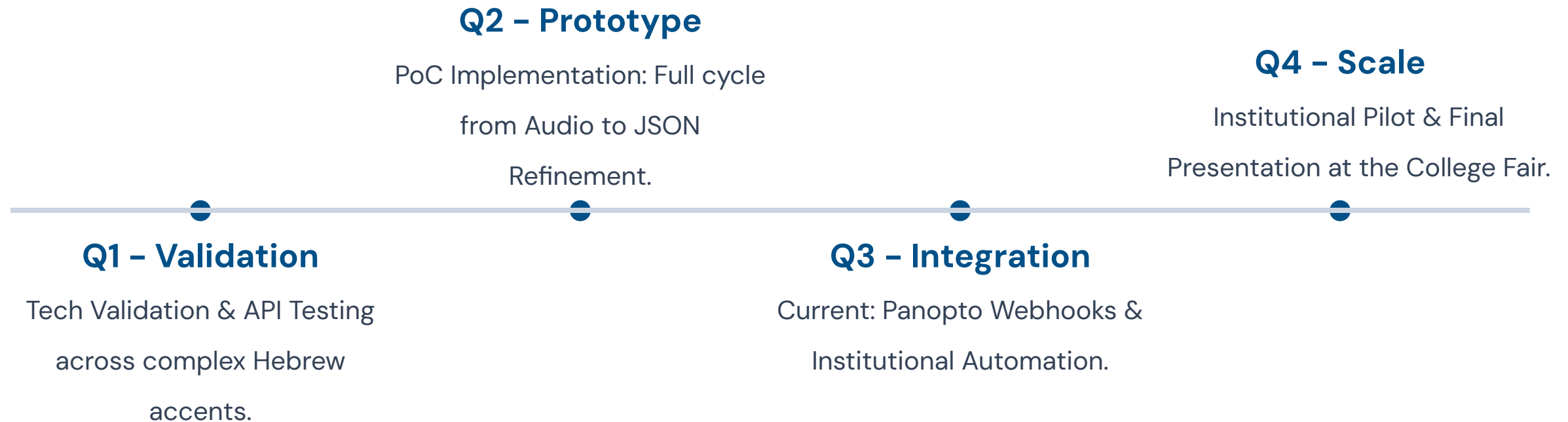
IDEAL CHOICE

Accessible Academic

- ✓ Specialized Hebrew STT
- ✓ Academic Terminologies
- ✓ Instant AI Study Materials
- ✓ Low-cost



Roadmap: From PoC to Institutional Scale



Questions?

Building a More Inclusive Academic Future, Together.

 <https://github.com/adi0258/AccessibleAcademic>

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